

Calculus mod 5

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About

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Lecture 1

What is algebraic geometry?

1.1 Introduction

Algebraic geometry is the study of shapes defined by systems of polynomial equations.

What makes it interesting to single out the class of shapes defined by polynomial equations in particular? In other words – why is algebraic geometry its own subject, and not just a source of examples for algebraic topology and Riemannian geometry courses?

There are several answers one could give to this question. The point of view taken by this course is that these shapes are interesting because they make sense over many different number systems.

To illustrate what we mean, let's single out the parabola $y = x^2$. There are many different ways one could interpret it:

1. if x, y are interpreted as *real numbers*, we get the standard parabola shape in $\mathbb{R}^2$;
2. if x, y are *complex numbers*, then $y = x^2$ defines a certain 2-dimensional subset of the 4-dimensional space $\mathbb{C}^2$;
3. if x, y are interpreted as *integers modulo 3*, then the parabola is the 'shape' consisting of the three points $(0, 0)$, $(1, 1)$, and $(2, 1)$;
4. if x, y are interpreted as *integers*, then the parabola is essentially the set of all perfect squares: it contains points like $(5, 25)$, $(13, 169)$, and $(-3, 9)$, where the x -coordinate can be any integer, and y is always its square.

The miracle of algebraic geometry is that these different incarnations of the shape $y = x^2$ are somehow related. The purpose of algebraic geometry, from this point of view, is to develop a language with which one can talk about all of these different incarnations at once, and therefore reason about the relationships between them.

Example 1.1.1 (Integrals and Diophantine equations). A high school calculus student can integrate

$$\int \frac{dx}{\sqrt{1-x^2}}. \quad (1.1.2)$$

However, the similar looking integral

$$\int \frac{dx}{\sqrt{x^3-2}} \quad (1.1.3)$$

would give even an expert undergraduate calculus student trouble!

When approaching the first integral, it is tempting to substitute $x = \sqrt{1-y^2}$. Then x and y obey the algebraic relationship

$$x^2 + y^2 = 1.$$

This allows you to evaluate (1.1.2) in terms of trigonometric functions (in fact, (1.1.2) evaluates to $\arcsin(y) + C$).

Similarly, the second integral is related to the algebraic curve

$$y^2 = x^3 - 2.$$

These two curves, $x^2 + y^2 = 1$ and $y^2 = x^3 - 2$, are related to the Diophantine equations

$$a^2 + b^2 = c^2 \quad (1.1.4)$$

and

$$a^2c = b^3 - 2c^3. \quad (1.1.5)$$

It's very easy to find (with proof!) all the integer solutions to (1.1.4); this was done by the ancient Greek mathematician Euclid; and even before Euclid, the mathematician Diophantus found an infinite family of solutions, and the Babylonians found many particular solutions.

However, (1.1.5) is a much trickier Diophantine equation. It has essentially one solution (observe $25 = 27 - 2$), but proving this is quite tricky unless you know some algebraic number theory!

Perhaps remarkably, the geometry of $x^2 + y^2 = 1$ explains why (1.1.2) is easy to solve, and why (1.1.4) is easy to solve; and the geometry of $y^2 = x^3 - 2$ explains why (1.1.3) and (1.1.5) are both hard to solve.

Example 1.1.6 (Dimension and point counts). Consider the parabola $y = x^2$. Over the real numbers, the graph of $y = x^2$ gives a 1-dimensional object. And in modulo n arithmetic, there are exactly n pairs (x, y) which solve

$$y \equiv x^2 \pmod{n},$$

as x can be any residue modulo n , and then y is determined.

Consider now the polynomial equation

$$x + y + z = 0.$$

Over the real numbers, this defines a 2-dimensional plane in 3-dimensional space. And in modulo n arithmetic, there are exactly n^2 solutions to $x + y + z \equiv 0 \pmod{n}$.

Note that, in both cases, the real number incarnation of our shape had dimension d , and the ‘modulo n ’ incarnation of our shape was a set of size n^d . The geometric notion of dimension is thus reflected arithmetically by the size of the solution set in modular arithmetic.

Remark 1.1.7. Dimension, intuitively, measures the ‘number of free parameters’ in a shape. If we are selecting d independent numbers in $\mathbb{Z}/n\mathbb{Z}$, then there are n^d total ways to make our selection, which explains why n^d corresponds to d -dimensions.

Example 1.1.8 (A more complicated dimension and point count example). Consider the equation

$$x^2 + y^2 = 1. \quad (1.1.9)$$

Over $\mathbb{R}$, the solutions to (1.1.9) form a circle – a 1-dimensional shape.

How many solutions are there modulo n ? Define

$$N_n := \text{number of solutions to } x^2 + y^2 \equiv 1 \pmod{n}.$$

Determining N_n is a fun exercise in elementary number theory¹. When $n = p^k$ is the power of a prime number p , like $n = 289 = 17^2$ or $n = 243 = 3^5$, then N_n is easy to determine:

$$N_{p^k} = \begin{cases} p^k - p^{k-1} & p \equiv 1 \pmod{4}, \\ p^k + p^{k-1} & p \equiv 3 \pmod{4}, \\ 2p^k & p = 2, k > 1 \end{cases}.$$

From this, using the Chinese remainder theorem, we can deduce that $N_n = O(n)$; here, we’re using big- O notation, meaning that N_n grows roughly as fast as n (and in particular, it grows slower than n^2 or n^3). This $O(n)$ is again related to the fact that the circle is 1-dimensional.

Example 1.1.10 (Point counting on the sphere). In the vein of the previous example, consider the equation

$$x^2 + y^2 + z^2 = 1.$$

Over the real numbers, this defines the unit sphere, a 2-dimensional shape. One can use elementary number theory, similarly to the previous example, to show that the number of solutions to

$$x^2 + y^2 + z^2 \equiv 1 \pmod{n}$$

is $O(n^2)$. For example, when $n = p^k$ where $p \equiv 1 \pmod{4}$ is a prime number, then there are $p^{2k} - p^{2k-1} = \left(1 - \frac{1}{p}\right) n^2$ solutions. The 2 in this $O(n^2)$ reflects the 2-dimensional nature of the solution set.

¹Hint: Use Legendre symbols!

1.2 Prespaces

To help formalize these ideas, we introduce now the notion of a *prespace*. A prespace is to algebraic geometry as a point-set topological space is to geometric topology: that is, all the objects of (classical) algebraic geometry are prespaces, but in practice algebraic geometers usually do not study arbitrary prespaces, and instead study prespaces obeying many extra properties. Compare this to how a geometric topologist does not study arbitrary point-set topological spaces, but usually assumes their spaces are Hausdorff, or metrizable, or locally Euclidean, etc.

In the introduction, we explained that the main benefit of studying shapes cut out by systems of polynomial equations is how they are realized over many different number systems. We are going to define a *prespace* as a shape which has an incarnation over every possible number system, where by number system we will mean *ring*.

Warning 1.2.1. In these notes, and in most notes on algebraic geometry, a ring is always assumed to be *commutative and unital*.

Definition 1.2.2. A *prespace* is a functor

$$X : \text{Ring} \rightarrow \text{Set}.$$

Remark 1.2.3 (What is a functor?). For those who do not know category theory, when we say $X : \text{Ring} \rightarrow \text{Set}$ is a *functor*, we mean that,

1. for every ring A , we can produce a *set* $X(A)$, and
2. for every ring *homomorphism* $A \rightarrow B$, we can produce a *function* of sets $X(f) : X(A) \rightarrow X(B)$,

such that, whenever $f : A \rightarrow B$ and $g : B \rightarrow C$ are two ring homomorphisms, we have $X(g \circ f) = X(g) \circ X(f)$.

Remark 1.2.4. It is useful to think of mathematical definitions as consisting of two parts: data, and then properties which that data must obey. For example, a group consists of the data of a set and a binary operation, obeying certain properties (like associativity).

Similarly, the data of a functor is the sets $X(A)$ and the functions $X(f)$, and the property this data must obey is $X(g \circ f) = X(g) \circ X(f)$.

Definition 1.2.5. A *morphism* of prespaces

$$\eta : X \rightarrow Y$$

is a natural transformation of functors.

Remark 1.2.6 (What is a natural transformation?). A morphism of prespaces is the data of, for every ring A , a function

$$\eta(A) : X(A) \rightarrow Y(A),$$

such that, for any ring homomorphism $\varphi : A \rightarrow B$, the diagram

$$\begin{array}{ccc}
X(A) & \xrightarrow{\eta(A)} & Y(A) \\
\downarrow X(\varphi) & & \downarrow Y(\varphi) \\
X(B) & \xrightarrow{\eta(B)} & Y(B)
\end{array}$$

commutes; that is, $Y(\varphi) \circ \eta(A) = \eta(B) \circ X(\varphi)$.

Remark 1.2.7. The reader should check that the following statements:

1. if $\eta : X \rightarrow Y$ and $\theta : Y \rightarrow Z$ are morphisms of prespaces, then the map $\theta \circ \eta : X \rightarrow Z$ defined by $(\theta \circ \eta)(A) := \theta(A) \circ \eta(A)$ is also a natural transformation;
2. a morphism of prespaces $\eta : X \rightarrow Y$ has an inverse $\eta^{-1} : Y \rightarrow X$ if, and only if, for every ring A , the function $\eta(A)$ is a bijection of sets.

Example 1.2.8 (The parabola). We define the *parabola* as the prespace

$$P : \text{Ring} \rightarrow \text{Set}$$

given by

$$P(A) := \{(x, y) \in A \times A \mid y = x^2\}.$$

Given a ring homomorphism $\varphi : A \rightarrow B$, we define

$$P(\varphi) : P(A) \rightarrow P(B)$$

as the function

$$P(\varphi)(x, y) = (\varphi(x), \varphi(y)).$$

This pair $(\varphi(x), \varphi(y))$ belongs to $P(B)$ because, whenever $y = x^2$, we have

$$\varphi(y) = \varphi(x^2) = \varphi(x \cdot x) = \varphi(x) \cdot \varphi(x) = \varphi(x)^2,$$

as φ is a ring homomorphism.

Example 1.2.9 (Affine space). We define *affine n -space* as the prespace

$$\mathbb{A}^n : \text{Ring} \rightarrow \text{Set}$$

given by

$$\mathbb{A}^n(A) := \{(x_1, \dots, x_n) \in A^n\}.$$

1.3 Affine schemes

We now single out a very special class of prespaces.

Definition 1.3.1. Let R be a ring. The *affine scheme* associated to R is the prespace

$$\text{Spec}(R) : \text{Ring} \rightarrow \text{Set}$$

given by

$$\text{Spec}(R)(A) := \{\text{ring homomorphisms } \varphi : R \rightarrow A\}.$$

Example 1.3.2. Consider the ring $\mathbb{Z}[x_1, \dots, x_n]$ of all polynomials in the variables $x_1, \dots, x_n$. Then the affine scheme $\text{Spec}(\mathbb{Z}[x_1, \dots, x_n])$ coincides with the prespace $\mathbb{A}^n$ of Example 1.2.9; that is, there is an isomorphism of prespaces

$$\eta : \text{Spec}(\mathbb{Z}[x_1, \dots, x_n]) \simeq \mathbb{A}^n.$$

This isomorphism η is defined as follows. For a ring A , set

$$\eta(A) : \text{Spec}(\mathbb{Z}[x_1, \dots, x_n])(A) \rightarrow \mathbb{A}^n(A)$$

the function sending a ring homomorphism $\varphi : \mathbb{Z}[x_1, \dots, x_n] \rightarrow A$ to the n -tuple $(\varphi(x_1), \dots, \varphi(x_n)) \in \mathbb{A}^n(A)$.

We leave it to the reader to check that η defines a morphism of prespaces. To see it is an isomorphism (in light of Remark 1.2.7), it is enough to check that $\eta(A)$ is always a bijection; we leave this to the reader (it is the universal property of polynomial rings).

Remark 1.3.3. The parabola P of Example 1.2.8 is also an affine scheme:

$$P = \text{Spec}(\mathbb{Z}[x, y]/(y - x^2)).$$

Can you see why?

Remark 1.3.4. Can you describe the set of ring homomorphisms

$$\mathbb{Z}[x, y, z]/(x^2 + y^2 + z^2 - 1) \rightarrow A,$$

for A an arbitrary ring, concretely?

Lecture 2

Tangent vectors and Hensel's lemma

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7